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New electronically fine-tunable CMOS floating resistor configuration and its application

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ABSTRACT

A new CMOS circuit configuration has been presented which can realize electronically fine-tunable linear voltage-controlled positive as well negative floating resistors from the same configuration by interchanging some circuit connections. The circuit employing eighteen MOSFETs, all of which are operated in saturation region, and the realised a floating resistance value is independent of the threshold voltage (V_{TH}) of the MOSFETs. The realised resistance has been found to vary from 71.26 k Ω to 35.73 k Ω for positive values and from -71.06 k Ω to -35.53 k Ω for negative values, when the control voltage is varied from 0.4 V to 0.8 V in both the cases. An application of the proposed circuit is demonstrated by using it in a new configuration of a first order current-mode universal filter based on unity gain current followers which is capable of realising low-pass, high-pass and all pass-all the three filters from the same configuration. The workability of the proposed voltage-controlled resistance (VCR) circuit in positive and negative both modes, as well that of the quoted application circuit, has been demonstrated through simulations using CADENCE SPICE tools and 0.18 μm CMOS technology parameters, at DC bias voltages of ± 1.25 V.

1. Introduction

Fully integratable systems in CMOS VLSI technology are although dominated by digital signal processing circuits; there are invariably several functions which may be required to be performed by CMOS analog circuits. Thus, in spite of the dominance of digital circuits and techniques, a number of functional circuits such as amplifiers, filters, rectifiers, A/D, and D/A converters, invariably require analog circuits. Research in CMOS analog circuits is, thus, motivated by the requirement of integrating both analog and digital parts on the same chip, using the same technology. Consequently, over the past five decades, a variety of CMOS compatible circuits such as switched-capacitor circuits, MOS-C circuits and translinear/square root domain circuits have been evolved to meet this requirement. This paper is concerned with the realization of CMOS positive/negative voltage-controlled floating resistors which are useful for the design of electronically-controllable analog filters, oscillators and waveform generators. In view of the above and to put the contribution of this paper in the right perspective, therefore, a brief account of some prominent CMOS linear voltage-controlled resistor (VCR) circuits evolved earlier is now in order.

The first floating linear CMOS VCR was proposed by Banu and Tsi-

vidis in [1]. This circuit uses a method to cancel out first order non linearity of the MOS transistor by bringing two of them in a parallel connection to devise a linear voltage-controlled floating resistor. Han and Pork [2] presented a new design of VCR using an unbalanced op-amp-based integrator configuration along with a matched pair of MOS transistors. Later, Singh, Hanson and Vlach [3] proposed a floating resistor using a non-linear cancellation technique implemented using current mirror. In next few years, Wang [5,6] proposed the design of floating VCRs based on square-law characteristic of the MOSFET, employing four matched transistors and two equal current sources. Sakurai and Ismail [7] presented a threshold voltage (V_{TH})-independent linear VCR. In [8], a low component-count implementation of floating VCR was proposed by Senani using op-amp-based scheme for cancelling the nonlinearity of the FET. Elwan, Mahmoud and Soliman [9] presented a floating VCR using fifteen MOSFETs with a minimum dependence on V_{TH} and thus, making the circuit immune against process and temperature variations. In [10], Mahmoud, Elwan and Soliman presented a new general method to implement CMOS floating VCRs, which resulted in two new configurations in which the magnitude of the floating resistance realised is controlled by an external voltage. Oura, Yoneyama, Tantry and Asai [11] proposed a voltage-controlled floating

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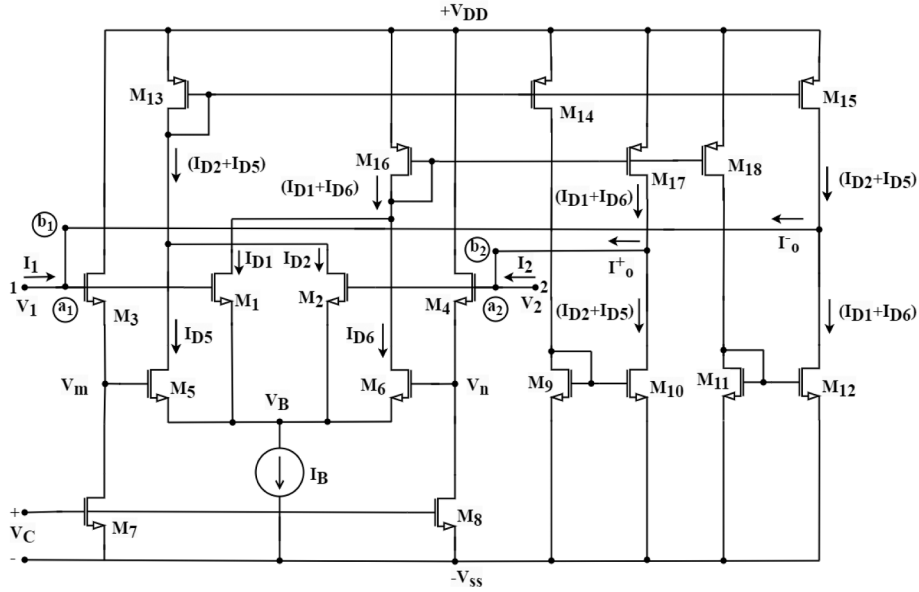


Fig. 1. The proposed positive/negative, floating, linear, voltage-controlled resistance (VCR) configuration.

resistor capable of operating for both positive and negative resistance modes, with independence on device threshold voltage. The application of low voltage floating resistor was investigated by Tantry, Ercan and Alci [12], in neuro-signal generation. In [13], Zouaoui-Abouda and Fabre presented high-valued floating VCR using two second generation current conveyors. Manolescu and Popa [14] used FGMOS transistor and linearization techniques to devise a CMOS active resistor circuit which was claimed to attain improved linearity. For low voltage applications, a low voltage CMOS current controlled floating negative resistor was proposed by Hamdi Ercon [16] which uses a self cascode composite transistor and offers a wide tunable range. A differential pair was exploited to design a low voltage floating resistor, employing a current controlling mechanism, by Tekin, Ercon and Alci [17]. The bias dependence of VCR was claimed to be eliminated by Yucel and Yuce [18] in their grounded voltage-controlled positive resistor design which has stringent matching requirement between active elements. Some recent floating positive and negative VCR designs by Senani, Rai, and Singh in [19–21] have been obtained by novel extensions of an earlier configuration reported by Yuce, Minaei and Alpaslan in [15]. However, the mentioned circuits require two control voltages $\pm V_C$ (of equal and opposite polarities). On the other hand, Chhabra, Senani, and Aggarwal [22,23] presented wide-range current-controlled positive floating VCR resistors using CMOS based translinear elements, however, the resulting circuit employ a considerably large number (as many as 64) of MOSFETs than the other approaches described above. A detailed account of the work related to electronically-tunable resistors has been provided by Senani, Bhaskar, Singh, and Singh in [24]. Yuce, Minaei and Herencsar [25] have introduced grounded positive VCR comprised of only five MOSFETs, which has ultra-low power consumption of 176 nW. A new wide band electronically tunable grounded VCR was subsequently introduced by Yuce, Tokat and Yucel [26] which consists of only 3-MOSFETs, does not require additional bias current or voltages and has only one controlled voltage.

From the above brief survey, it is seen that the various CMOS linear VCRs known earlier have different features and exhibit different performance. More recently, there has been some interest on electronically fine-tunable circuits, for instance, see [18,32,38] which, although may not possess very wide tuning range of the parameters but nevertheless, are useful in applications where fine-tunability of certain parameters will suffice.

The main object of this paper is to present a new electronically fine-

tunable CMOS linear VCR configuration which can realise both positive and negative floating VCRs from the same circuit by the simple artifice of changing some circuit connections. By grounding either of the two parts, the same circuit can also realise positive/negative grounded VCRs. The application of the proposed circuit in floating positive VCR mode has been demonstrated in realising a unity-gain current followers-based electronically fine-tunable first order current-mode universal active filter from which low-pass, high-pass and all-pass filter may be realized from the same configuration. The workability of the proposed configuration, in its various modes as well as the application example of first order current-mode universal filter, have been verified by Cadence SPICE simulations and some results have been given to demonstrate the validity of the new propositions.

2. Proposed electronically fine-tunable CMOS VCR

The proposed CMOS VCR circuit is shown in Fig. 1. The genesis of this circuit is inspired by the work of Wang and Guggenbuhl [4].¹ It is assumed that the MOSFETs M_1 - M_8 , M_9 - M_{12} and M_{13} - M_{18} are matched and all are operated in saturation. Since M_3 and M_7 are matched and carry equal drain currents, it follows that their gate to source voltages would be equal. Hence, $V_1 - V_m = V_C$. The same argument applies to the MOSFETs M_4 and M_8 due to which $V_2 - V_n = V_C$, where V_C is the external control voltage. Now, by a straight forward analysis of the circuit, the port 1 current can be expressed as:

$$I_1 = -I_o^- = -\{(I_{D2} + I_{D5}) - (I_{D1} + I_{D6})\} \quad (1)$$

By expressing the various drain currents by the square law, Eq. (1) can be expanded as:

$$I_1 = -\{K(V_2 - V_B - V_{TH})^2 + K(V_m - V_B - V_{TH})^2 - K(V_1 - V_B - V_{TH})^2 - K(V_n - V_B - V_{TH})^2\} \quad (2)$$

where $K = \frac{1}{2}\mu_n C_{ox} \frac{W}{L}$ and the symbols having their usual meaning (for

¹ An alternative circuit with the same number of MOSFETs is feasible by following [57], with the current source I_B deleted and instead, the source terminals of the MOSFETs M_1 , M_2 , M_5 and M_6 –all connected to negative power supply $-V_{SS}$.

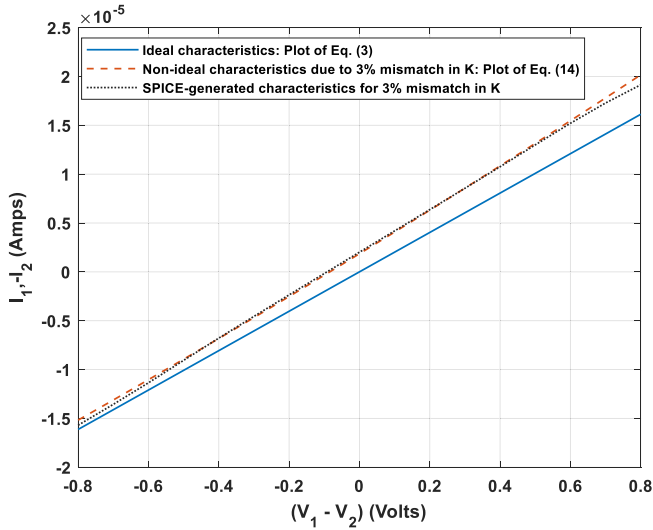


Fig. 2. Plot of Eq. (3), non-ideal Eq. (14) with 3% mismatch and SPICE-generated characteristics for 3% mismatch in K.

instance, see [4]). After algebraic simplifications and noting that $(V_m - V_n) = (V_1 - V_2)$, Eq. (2) reduces to the following:

$$I_1 = 2KV_C(V_1 - V_2) \quad (3)$$

Similarly, port 2 current of the circuit of Fig. 1 can be written as:

$$I_2 = -I_O^+ = -\{(I_{D1} + I_{D6}) - (I_{D2} + I_{D5})\} \quad (4)$$

By putting the values of the various drain currents using the square law,

$$I_1 = -I_2 = -I_O^- = 2KV_C(V_1 - V_2) + (\Delta K_1 V_1^2 - \Delta K_5 V_m^2) - (\Delta K_2 V_2^2 - \Delta K_6 V_n^2) + 2(V_B + V_{TH})(\Delta K_2 V_2 - \Delta K_1 V_1 + \Delta K_5 V_m - \Delta K_6 V_n) + (V_B + V_{TH})^2(\Delta K_1 - \Delta K_2 + \Delta K_6 - \Delta K_5) \quad (11)$$

Eq. (4) can be expanded as:

$$I_2 = -\{K(V_1 - V_B - V_{TH})^2 + K(V_n - V_B - V_{TH})^2 - K(V_2 - V_B - V_{TH})^2 - K(V_m - V_B - V_{TH})^2\} \quad (5)$$

which, upon simplification, results in:

$$I_2 = -2KV_C(V_1 - V_2) \quad (6)$$

From Eq. (3) and Eq. (6), the Y-matrix for the circuit of Fig. 1 can be written as:

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = 2KV_C \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} \quad (7)$$

From Eq. (7), it follows that the circuit of Fig. 1 represents a floating positive VCR and the realised equivalent resistance of the proposed circuit is, therefore, given by:

$$R_{eq} = \frac{1}{2KV_C} \quad (8)$$

It is interesting to point out that if the node a_1 is connected to node b_2 and node a_2 connected to node b_1 , then the transformed circuit would realise a floating negative VCR. This can be easily verified by a similar re-analysis through which the admittance matrix of the negative VCR turns out to be:

Table 1

Aspect ratios of the various MOSFETs of the proposed VCR of Fig. 1.

MOSFETs	Aspect ratio $W(\mu m)/L(\mu m)$
$M_1, M_2, M_3, M_4, M_5, M_6, M_7, M_8$	0.72/4.14
$M_9, M_{10}, M_{11}, M_{12}$	0.72/2.16
$M_{13}, M_{14}, M_{15}, M_{16}, M_{17}, M_{18}$	18.5/2.16

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = -2KV_C \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} \quad (9)$$

From Eq. (9), the realised equivalent negative resistance of this alternative circuit is, therefore, given by:

$$R_{eq} = -\frac{1}{2KV_C} \quad (10)$$

From the Eq. (8) and Eq. (10), it follows that in both the cases, the realised resistance value is electronically tunable through the external control voltage V_C . Furthermore, it may be noted that in both the cases, the realised R_{eq} is threshold voltage (V_{TH}) –independent.

It may also be pointed out that, with either of the two ports grounded, the same circuit can also be used to realize positive/negative grounded VCRs.

3. Mismatch analysis

The core of the circuit of Fig. 1 is the sub-circuit made up of M_1, M_2, M_5 and M_6 . Hence, we consider the mismatch in K parameters of these MOSFETs only by taking $K_1 = K + \Delta K_1$, $K_2 = K + \Delta K_2$, $K_5 = K + \Delta K_5$ and $K_6 = K + \Delta K_6$. A re-analysis of the circuit of Fig. 1 reveals the following:

$$\text{If we take, } \Delta K_1 = \Delta K_2 = \Delta K_5 = \Delta K_6 = \Delta K, \text{ then Eq. (11) reduced to} \quad (12)$$

$$I_1 = -I_2 = 2(K + \Delta K)V_C(V_1 - V_2)$$

which is essentially of same form as Eq. (3) with K replaced by $(K + \Delta K)$.

If we take $\Delta K_1 = \Delta K_5 = \Delta K$ and $\Delta K_2 = \Delta K_6 = -\Delta K$, and define $V_d = (V_1 - V_2)$ and $V_{cm} = \frac{(V_1 + V_2)}{2}$ then, Eq. (11) reduces to

$$I_1 = -I_2 = 2KV_C(V_1 - V_2) + 4\Delta KV_C(V_{cm} - V_B - V_{TH}) \quad (13)$$

In this case too, the linear part remains same as in Eq. (3), but the second term indicates a common-mode component as well as a DC offset.

Lastly, if the mismatches are such that $\Delta K_1 = \Delta K_6 = \Delta K$ and $\Delta K_2 = \Delta K_5 = -\Delta K$, then the expression (11) takes the form:

$$I_1 = -I_2 = 2KV_C V_d + \Delta K(V_d)^2 + 4\Delta K(V_{cm} - V_B - V_{TH})^2 \quad (14)$$

Thus, it is seen that besides the first term, which is the intended one, the inputs V_1 and V_2 are appearing as squared in the second term (due to V_d) as well as in the third term (due to V_{cm}) and there is a DC offset current implicit in the third term. It may be noted that both V_d^2 and V_{cm}^2 are going to be small since input signal would be in mV and furthermore, their multiplying factors are ΔK and $4\Delta K$, respectively, both of which are small, hence, the contribution of the overall square nonlinearity would be very small, as compared to the intended linear (first) term. However, the same can not be said about the DC offset term.

In view of the above, therefore, to make a quantitative assessment of the contribution of the unwanted terms of Eq. (14), the ideal expression

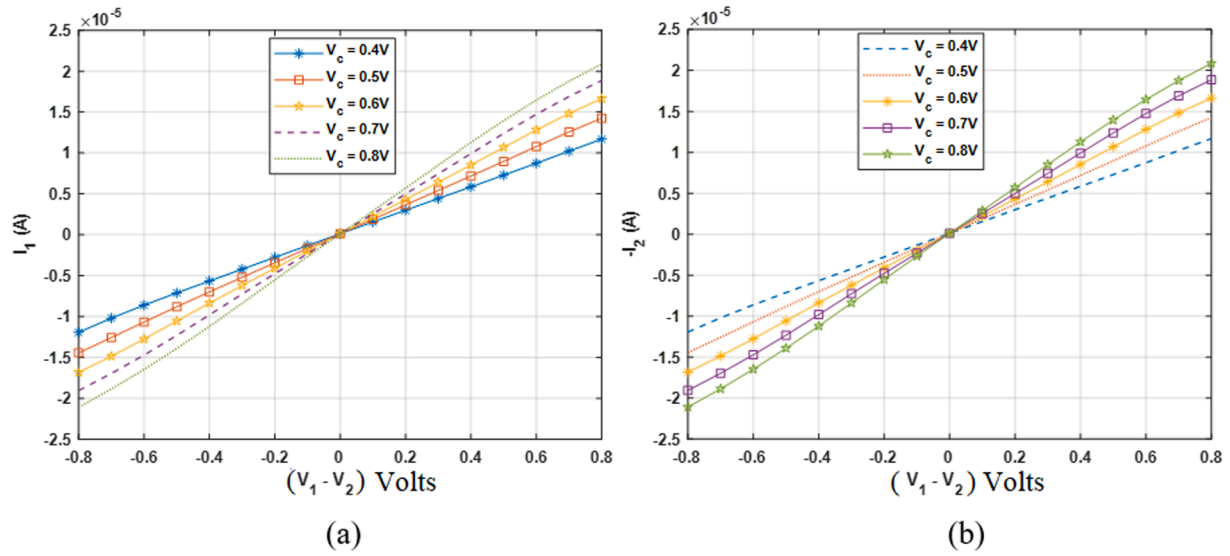


Fig. 3. DC Transfer characteristics of the circuit of Fig. 1 configured as a floating positive VCR, (a) I_1 as a function of $(V_1 - V_2)$, (b) $-I_2$ as a function of $(V_1 - V_2)$.

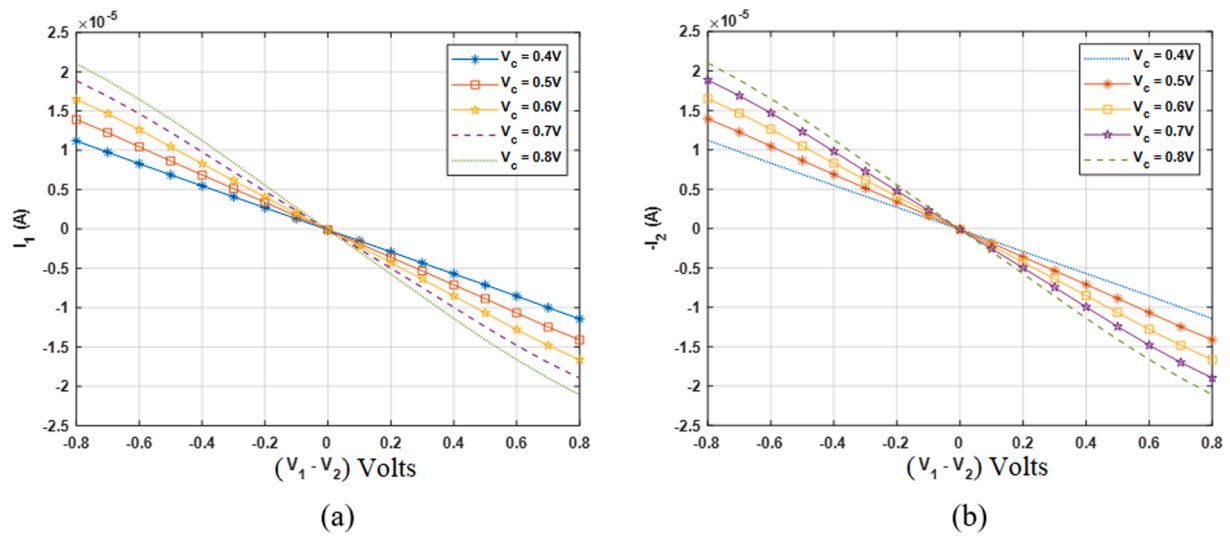


Fig. 4. DC Transfer characteristics of the circuit of Fig. 1 configured as a floating negative VCR, (a) I_1 as a function of $(V_1 - V_2)$, (b) $-I_2$ as a function of $(V_1 - V_2)$.

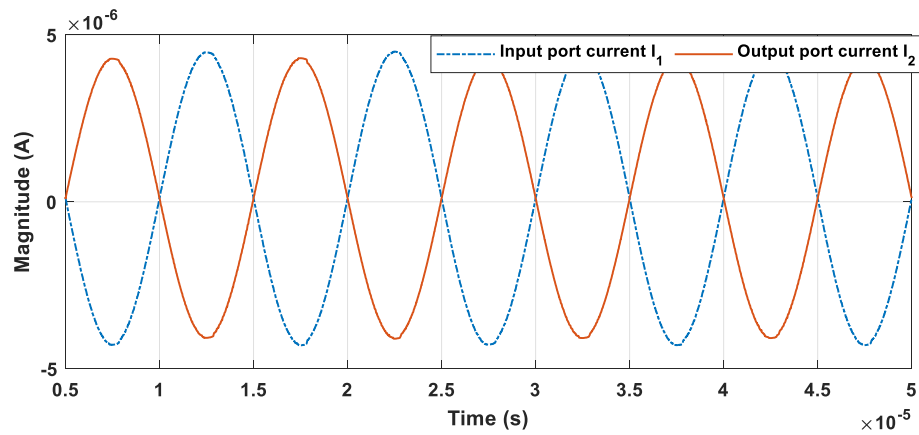


Fig. 5. Transient response of the floating, linear, positive voltage-controlled resistance (VCR) configuration.

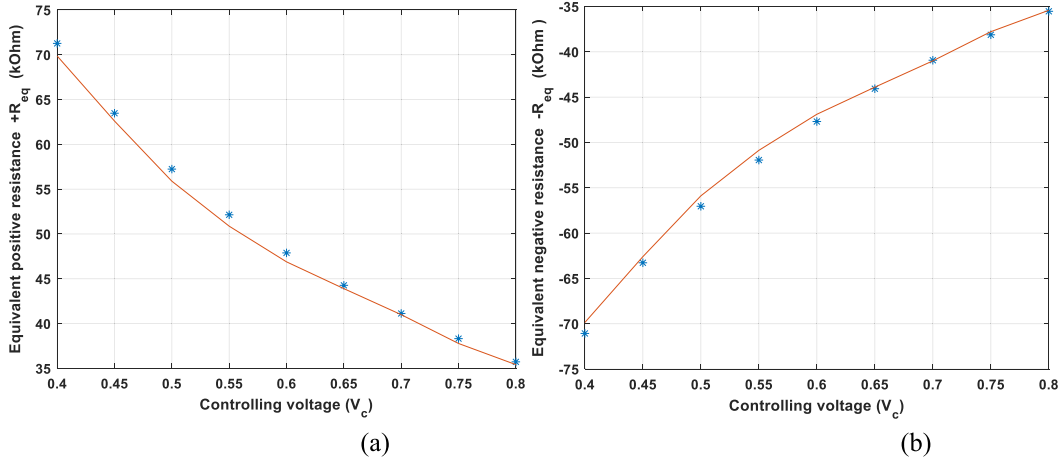


Fig. 6. Variation of the realised floating VCR as a function of the control voltage V_c for the circuit of Fig. 1 (a) for positive floating VCR (b) for negative floating VCR.

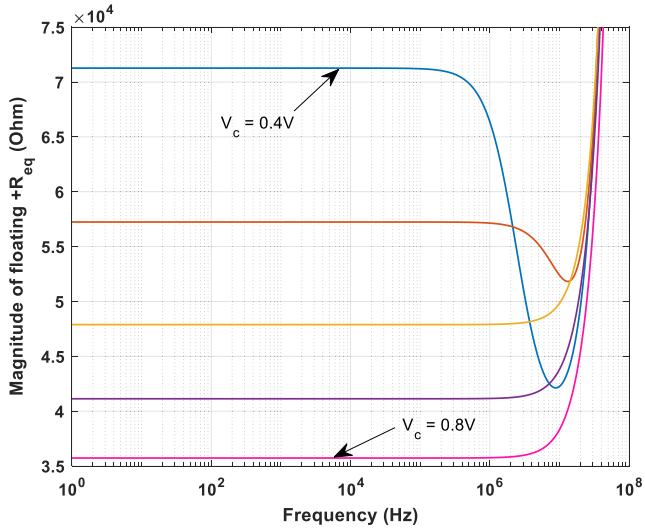


Fig. 7. Magnitude response of the floating positive VCR realised from the circuit of Fig. 1.

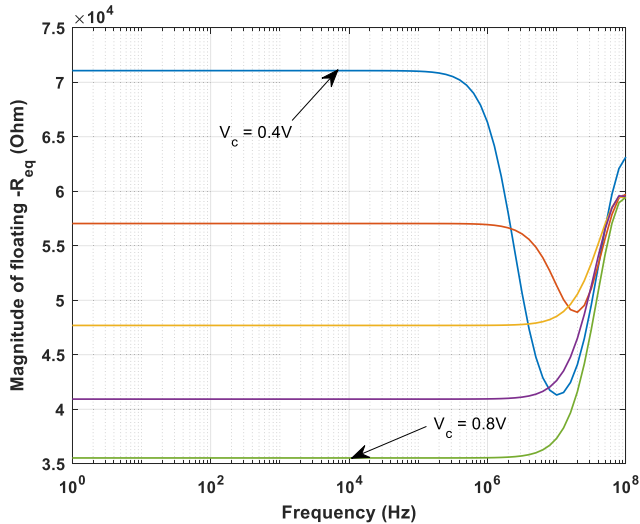


Fig. 8. Magnitude response of the equivalent resistance of the circuit of Fig. 1 as negative floating VCR.

of Eq. (3) and its non-ideal counterpart of Eq. (14) have been plotted in Fig. 2 by taking various parameter values (which are compatible with those taken in SPICE simulations in the next section) as: $K = 1.84 \times 10^{-5}$ amp/V² (corresponding to $\mu_s = 259.53 \text{ cm}^2/\text{V-s}$, $C_{ox} = 8.42 \text{ nF}/\mu\text{m}^2$ and $W/L = 0.72/4.14 \mu\text{m}$), $\Delta K = 0.055 \times 10^{-5}$ (3 % mismatch), $V_{TH} = 0.382 \text{ V}$, $V_B = -1.35 \text{ V}$, $V_C = 0.6 \text{ V}$ with V_d varied from -0.8 to $+0.8 \text{ V}$. SPICE-generated characteristics for the 3 % mismatch in K , conforming to the derivation of Eq. (14), has also been plotted, with the same parameters. It is observed from the plots of Fig. 2 that the effect of nonlinearity is not observable due to the contribution of the square terms of Eq. (14) being very small but a DC offset is, however, observable which is found to be 1.96×10^{-6} amp at $V_d = 0$.

4. Performance Evaluation through SPICE simulations

The functionality of the proposed circuit has been checked through various simulations using Cadence SPICE employing $0.18 \mu\text{m}$ CMOS technology parameters. The aspect ratios of the various MOSFETs have been taken as shown in Table 1. The power supply voltages are taken as $\pm 1.25 \text{ V}$ and the DC bias current was taken as $I_B = 40 \mu\text{A}$ (found to be the best choice after a number of trials).

Both port currents I_1 and $-I_2$ are plotted as a function of $(V_1 - V_2)$ and are depicted in Fig. 3 for different values of the control voltage. The characteristics of Fig. 3 show the nearly similar plots of I_1 and $-I_2$ which confirms the realisability of a floating resistance in both the cases, a good linearity of the realised positive VCR over a range of input voltage varying from -0.8 V to $+0.8 \text{ V}$ has been obtained. The realised resistance value varies from $71.26 \text{ k}\Omega$ to $35.73 \text{ k}\Omega$ as the control voltage has been changed from 0.4 V to 0.8 V .

The DC transfer characteristics of the proposed circuit in floating negative VCR mode are shown in Fig. 4. It has been observed that the negative resistance value varies from $-71.06 \text{ k}\Omega$ to $-35.53 \text{ k}\Omega$ as the control voltage is changed from 0.4 V to 0.8 V .

The transient analysis of the circuit was carried out by applying an input sinusoidal signal of amplitude 200 mV (p-p) at frequency of 100 kHz , with port 2 shorted and then monitoring the port currents I_1 and I_2 which are shown in Fig. 5 and are seen to be in phase opposition, as expected. From Fig. 5, it has been observed that for control voltage $V_C = 0.6 \text{ V}$ (resulting in $R_{eq} = 47.88 \text{ k}\Omega$), both the port currents of the proposed circuit have been found to be $4.18 \mu\text{A}$.

Fig. 6 represents the variation of realised simulated and theoretical resistances in both the cases, on varying the control voltage V_c from 0.4 V to 0.8 V . The realised value of positive resistance was found to vary from $71.26 \text{ k}\Omega$ to $35.73 \text{ k}\Omega$ whereas the negative resistance was found to vary from $-71.06 \text{ k}\Omega$ to $-35.53 \text{ k}\Omega$.

Fig. 7 and Fig. 8 show the frequency responses of the equivalent

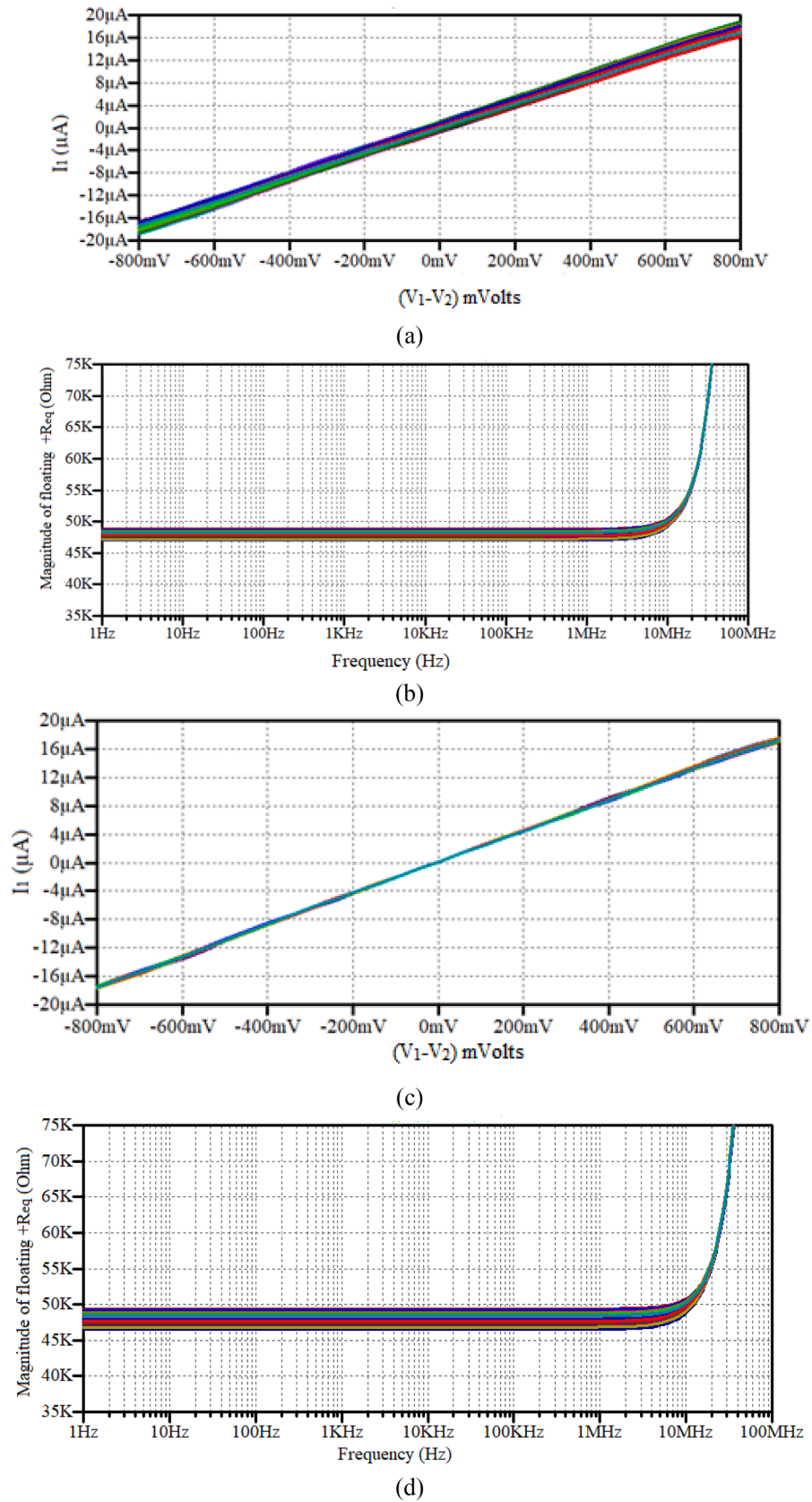


Fig. 9. Monte Carlo analysis, (a) V-I characteristics I_1 as a function of $(V_1 - V_2)$ for 3% mismatch in aspect ratios of the transistors, (b) Frequency response for 3% mismatch in aspect ratios of the transistors, (c) V-I characteristics I_1 as a function of $(V_1 - V_2)$ for 5% mismatch in the threshold voltage of transistors, (d) Frequency response for 5% mismatch in the threshold voltage of the transistors.

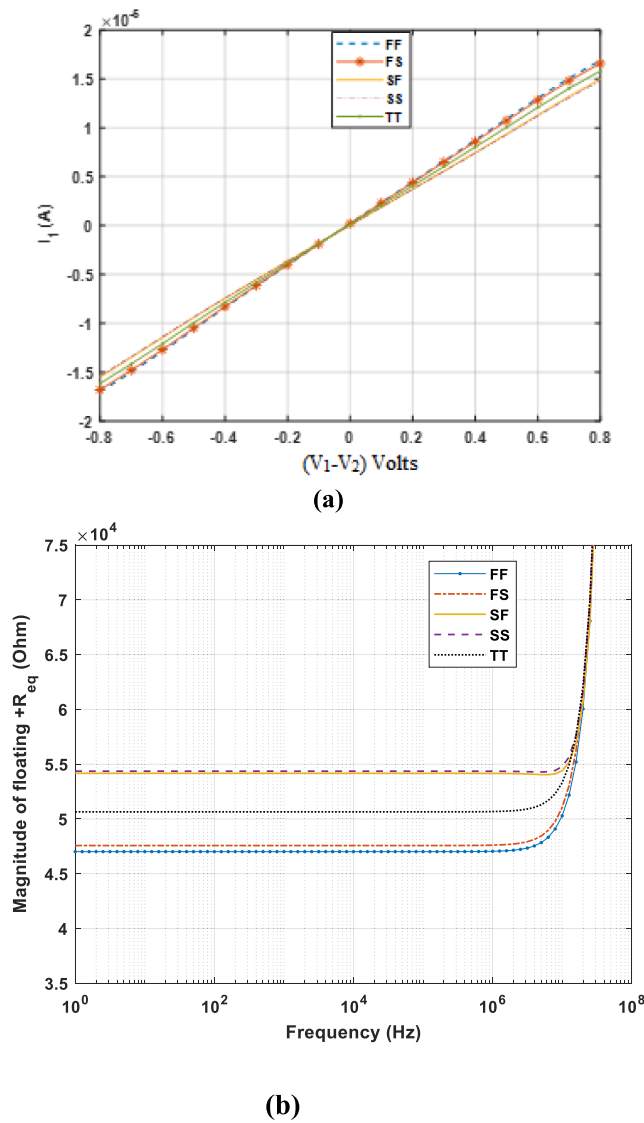


Fig. 10. Corner analysis of the circuit of Fig. 1 (a) V-I characteristics for I_1 as a function of (V_1-V_2) (b) Magnitude response of equivalent floating positive resistance.

resistance of the positive and negative VCR modes of the proposed circuit. From Fig. 7, it has been observed that the frequency response of the proposed positive floating VCR is almost flat for the frequency ranging from 930.45 kHz to 10.83 MHz. if the control voltage is varied from 0.4 V to 0.8 V. Nearly similar variations can be seen for the negative floating VCR, as depicted in Fig. 8.

To check the effect of the statistical changes of various transistor parameters, such as threshold voltage and aspect ratios on the performance of the circuits, a Monte Carlo simulation of the circuit of Fig. 1 for positive floating VCR was performed for $V_C = 0.6$ Volts. Fig. 9a and Fig. 9b show the effect of 3 % change in the aspect ratio of the transistors on the V-I characteristic and the frequency, while Fig. 9c shows the V-I characteristics of the proposed circuit for 5 % fluctuation in the threshold voltage of the transistors, for one hundred runs. The corresponding magnitude response is shown in Fig. 9d.

The results of Fig. 9 confirm that the effect of various statistical changes in the considered parameters is very small on the performance of the proposed circuit. It may be emphasised that in view of the V_{TH} independence of the realised resistance as reflected in Eq. (8) and Eq. (10), the least variations in the V-I characteristics of the proposed circuit are observable in case of the plot of Fig. 9c.

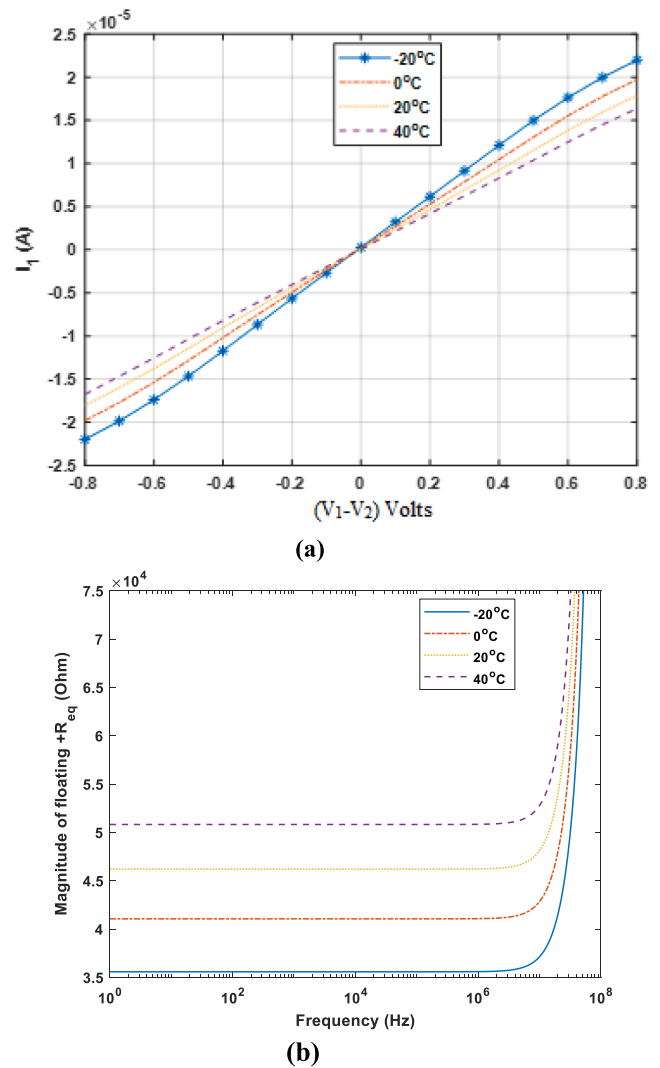


Fig. 11. Temperature analysis of the circuit of Fig. 1, (a) V-I characteristics for I_1 as a function of (V_1-V_2) (b) Magnitude response of realised floating positive resistance.

The Corner analysis of the new positive floating VCR of the circuit of Fig. 1 was also performed by taking different corner technology parameters fast-fast (FF), fast-slow (FS), slow-fast (SF), slow-slow (SS) and at typical-typical (TT) to check the changes in the circuit behaviour at different corner parameters. Fig. 10a shows the V-I characteristics of the circuit of Fig. 1 for all corner analyses. Fig. 10b shows the magnitude of equivalent floating positive resistance to be 1.24, 0.987, 0.977, 0.974 and 0.801 k Ω for SS, SF, TT, FS and FF, respectively. To check the effect of the temperature variation, the temperature analysis of the proposed circuit was also carried out. Fig. 11a show the V-I characteristics of the proposed circuit for different values of the temperature ranging from -20°C to 40°C in the steps of 20°C . The magnitude response of the realised floating resistance is shown in Fig. 11b.

Fig. 12 shows the layout of the proposed positive floating VCR of Fig. 1. To check the performance of the proposed circuit, post-layout simulations are carried out. Fig. 13a shows the V-I characteristics of the proposed circuit for the control voltage of 0.6 V. Fig. 13b shows the magnitude response of the equivalent resistance, from which it is observed that the resistance value of the proposed circuit can be found to be 52.14 k Ω against the nominal value of 47.89 k Ω . The area of the proposed floating VCR is found to be 1129.954 μm^2 ($x = 30.1 \mu\text{m} \times y = 37.54 \mu\text{m}$).

The comparative features of proposed circuit, with the previously

known FVCR configurations in which MOS transistors are operated in saturation region, are shown in Table 2. From the perusal of Table 2, it can be seen that some of the parameters of the proposed circuit are, although inferior to the previously published circuits, but on the other hand, other parameters are better or superior. Therefore, to take a comprehensive view and better comparison, we propose the following Figure of Merit (FOM):

$$\text{Figure of Merit (FOM)} = \frac{\text{Operating frequency range (MHz)} \times \text{Maximum applicable input voltage (mV)}}{\text{Power supply voltage} \times \text{No. of MOS transistors} \times \text{Power consumption (mW)}}$$

From Table 2 it can be seen that the FOM of the proposed circuit is, although smaller than the circuits of [16,17] and [18], but is far greater than those of [7,19] and [20].

5. Application: Electronically fine-tunable first-order universal current-mode filter

In this section, we demonstrate a novel application of the proposed circuit of Fig. 1 is realised a first order universal current mode filter. First-order universal filter is a multi-function filter circuit which enables the realization of low pass, high pass and all pass functions from the same circuit. Some previously known first-order universal filters, which are made using different building blocks, are given in [27–56] which can be summarised as follows:

Ozoguz, Toker, and Cicekoglu [27], introduced a first-order current-mode universal filter using three inverting type second generation current conveyors (ICCI), two resistors, and two capacitors. Later, Horng [28,29] devised universal filters based on one or two differential voltage current conveyor (DVCCs) and one or two resistors along with one capacitor. Khan, Simsim, and Beg [30] developed a universal filter using two digitally programmable current conveyors (DPCCII), one resistor,

and one capacitor. In order to add more features, Myderrizi, Minaei, and Yuce [32] presented a novel, electronically fine-tunable multi-input-single-output (MISO) universal filter, offering versatility in filter implementation. Ansari and Soni [33] presented a digitally programmable universal filter using one digitally controllable-differential voltage current conveyor (DC-DVCC), two resistors, and one capacitor. Senani, Bhaskar, and Singh in [34] have provided a comprehensive

overview of filter design using current conveyors. Yuce and Minaei [35] proposed a first-order current-mode universal filter using two dual-output second-generation current-controlled conveyors (DO-CCII), one

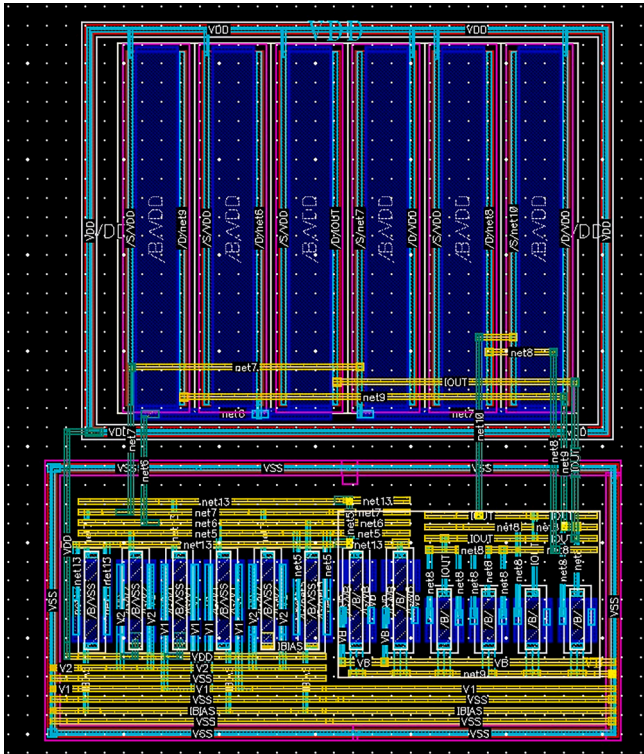
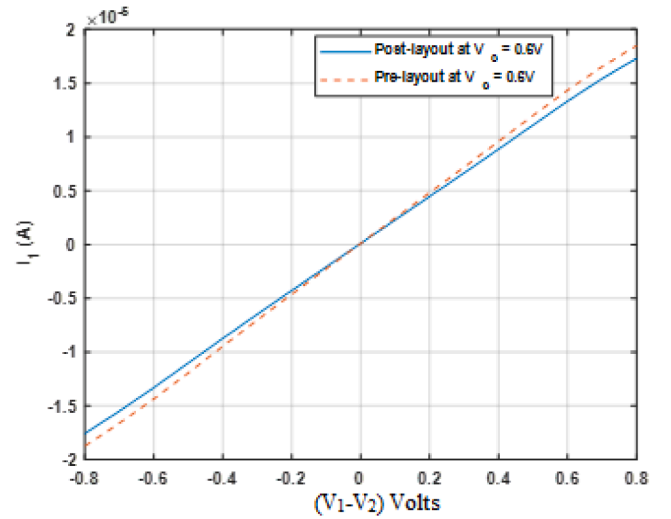
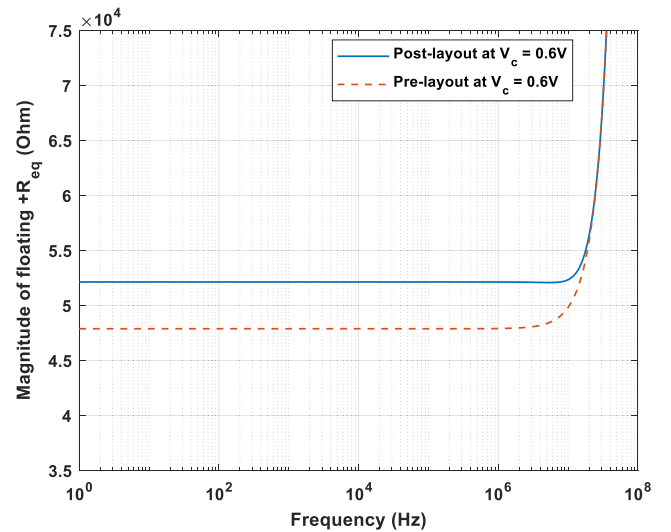


Fig. 12. Layout of the proposed circuit of Fig. 1 in positive floating VCR mode.



(a)



(b)

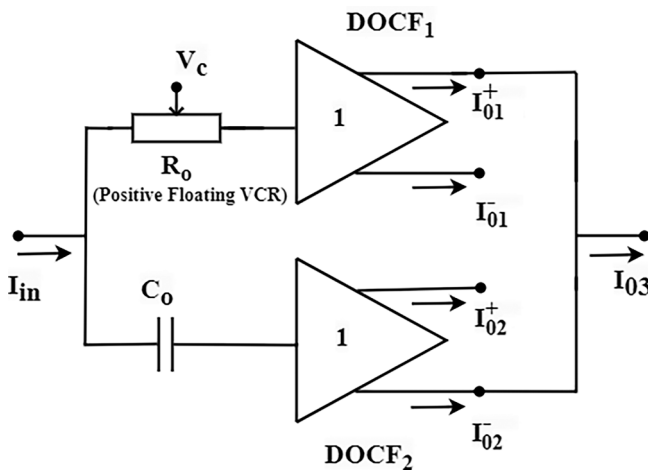
Fig. 13. Post-layout simulation results of the circuit of Fig. 1 (a) I_1 as a function of (V_1-V_2) , (b) Magnitude of floating resistance as a function of frequency.

Table 2

Comparative features of the previously known linear FVCR Configurations in which MOSFETs are operated in saturation mode.

Reference Number	Number of MOSFETs	Power Supply	Type of Resistance	Linearity Range	Power Consumption	Range of Resistance	Frequency Range of Operation (Hz)	Chip Area	Figure of Merits (FOM)
[1]	6	—	Positive	± 4 V	—	—	—	—	—
[3]	18, 36	± 5 V	Positive	± 1 V	—	—	—	—	—
[5]	6	5 V	Positive	4 V	—	—	—	—	—
[6]	4 + 2 CS	—	Positive	± 2 V	—	—	—	—	—
[7]	18	± 5 V	Positive	± 1.5 V	0.4–4 mW	85 k–65k Ω	10 M	210 $\mu\text{m} \times$ 270 μm	4.167
[9]	14,15	± 3.3 V	Positive	NA	Less than 0.5mW	—	—	—	—
[10]	18,20	± 3.3 V	Positive	± 0.75 V	Less than 1 mW, less than 1.33 mW	—	—	—	—
[11]	40	± 5 V	Both Positive and Negative	± 2.5 V	NA	1.7 M to ∞ –1.21 M to ∞	70 k	—	—
[16]	15 + 1CS	± 0.75 V	Positive	± 0.25 V	65 μ W	5.9 k to 243k	222.1 M	—	7593.165
[17]	15 + 1CS	± 0.75 V	Both Positive and Negative	± 0.5 V	0.41 m W	1.5 k to 1 M	1.66 G	—	17994.579
[19]	16	± 1.5 V	Both Positive and Negative	± 0.3 V	0.26 to 19.4 m W	–2.57 k to –0.389 k	10 M	—	3.468
						2.53 k to 0.375 k			
[20]	16	± 1.5 V	Both Positive and Negative	± 0.3 V	0.25 to 18.1 m W	–2.4 k to –0.448 k	50.23 M	—	0.644
						2.37 k to 0.431 k			
[23]	64	± 0.6 V	Positive	± 0.03 V	26 μ W	13k Ω to 1.3G Ω	50 M	—	150.24
Present work	18	± 1.25 V	Both Positive and Negative	± 0.8 V	0.30 mW	71.26 k Ω to 35.73 k Ω –71.06 k to –35.53 k	10 M	30.1 $\mu\text{m} \times$ 37.54 μm	118.52

(CS-current source; ‘-’: concerned information not available).

**Fig. 14.** An entirely CMOS-based first-order voltage-controllable current-mode universal filter.

resistor, and one capacitor. Safari, Yuce, and Minaei [36] introduced a resistorless, electronically-tunable universal filter employing two inverting second-generation current conveyors (ICCI), one MOSFET-based resistor, and one capacitor. Kumar and Paul [37] presented a first-order universal filter with a multiphase sinusoidal oscillator using one dual-X multiple-output second-generation current conveyor (DX-MOCCII), one resistor, and one capacitor. Masud, Aain, and Khan [39] proposed a universal filter utilizing two DC-DVCCs, three resistors, and three capacitors. In a voltage-mode approach, Abaci and Yuce [40] presented a first-order universal filter using two subtractors, one

resistor, and one capacitor, offering an alternative to current-mode designs with different trade-offs in impedance and power consumption. Chaturvedi, Kumar, and Mohan [41] introduced a first-order universal filter using one dual-X current conveyor transconductance amplifier (DX-CCTA) and two capacitors. Agarwal and Maheshwari [42] proposed a universal filter using one Extra-X current controlled conveyor (EX-CCCI) and one capacitor. Similarly, Chaturvedi, Mohan and Kumar [43] developed a current-mode filter with voltage-mode transformation capabilities using two multi-output second-generation current conveyors, a resistor, and a capacitor. Horng, Wu, Zheng and Li [44] employed two ICCIIs, one resistor, and one capacitor in their first order universal filter design. Chaturvedi, Mohan and Kumar [45] devised a resistor-less first-order universal filter with a single differential difference dual-X second-generation current conveyor (DD-DXCCII) and a capacitor. Yuce and Minaei [46] introduced a new first-order universal filter configuration using two plus-type inverting second-generation current conveyors (ICCI+), a resistor, and a grounded capacitor, offering various filtering applications with enhanced performance. Yucel [47] introduced a DVCC-based current-mode first-order universal filter. Dogan and Yuce [48] proposed a first-order universal filter circuit that uses two current feedback operational amplifiers (CFOAs), four resistors, and one capacitor, providing adjustable gain and low output impedance. Raj, Bhaskar, Senani, and Kumar [49] extended their earlier all-pass filter configurations to first-order universal active filters, demonstrating how advances in one filter type can enhance others. In [50], they introduced another first order universal filter circuit which uses only a single CFOA, one resistor, and one capacitor. Rohilla, Pushkar, Kumar, and Raj [51] introduced resistorless filter design using operational transconductance amplifiers (OTAs). Raj, Senani, and Bhaskar [52] presented a first-order universal filter design using a single-differential difference current conveyor, tailored for versatile voltage-mode and transadmittance-mode applications, highlighting its simplicity and efficiency. Recently,

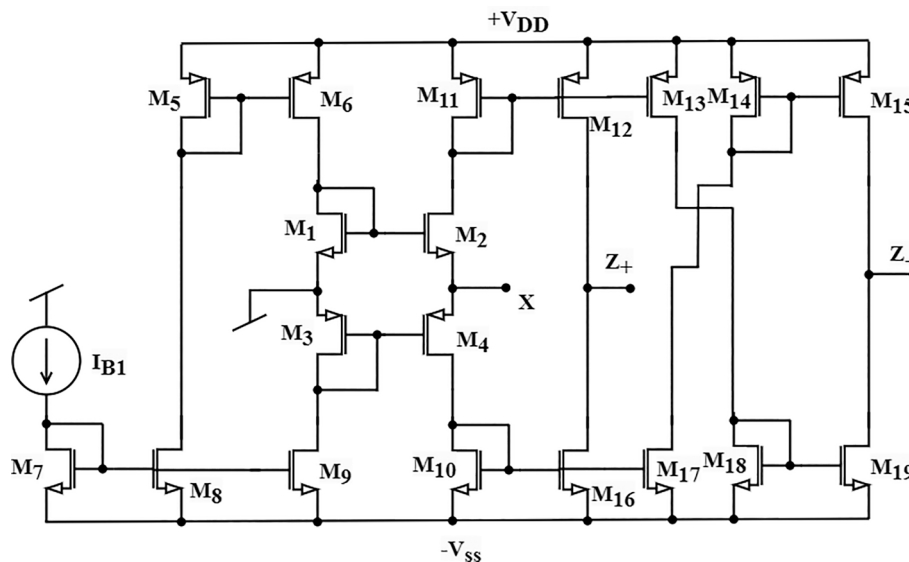


Fig. 15. An exemplary circuit implementation of the of DO-CF (based on the dual output CCII of [29]).

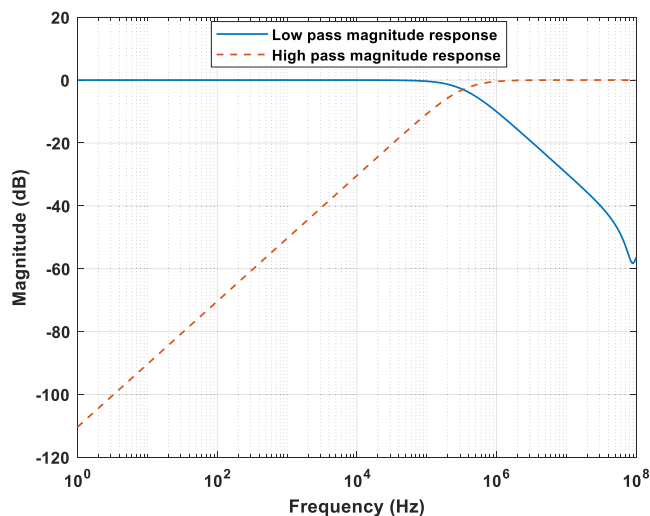


Fig. 16. Frequency response of LP and HP filter realised from the circuit of Fig. 14.

Khateb, Kumngern and Kulej [53] have presented a 0.5-V nano-power voltage-mode first-order universal filter based on multiple-input OTA, whereas Kumngern, Jongchanachavawat, Phatsornsiri, Wongprommoon, Khateb and Kulej [54] introduced a current-mode first-order versatile filter using Translinear Current Conveyors with controlled current gain. Subsequently, Khateb, Kumngern, Kulej, Stopjakova and Psychalinos [55] presented a voltage-mode first order analog filter using a single multiple-input voltage differencing differential difference amplifier (VDDDA) operated with still lower voltage of 0.3 V with power consumption as low as 357.4 nW. More recently, Kumngern, Khateb and Kulej [56] have come up with a single EX-CCCII-based first-order versatile active filter having a single input and four current outputs and realizing all the three first order filters, with the all-pass filter realisable with a positive as well negative polarity.

In the following, we present an application of the proposed finetunable active resistor, in realising a first order current mode universal filter, which has been implemented using two dual output current followers (DO-CF) and is shown in Fig. 14 whereas an exemplary CMOS based circuit of DO-CF is given in Fig. 15. We may mention that, to the

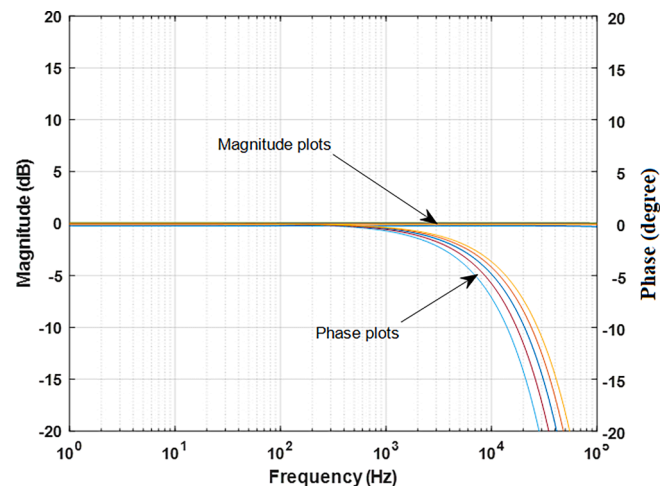


Fig. 17. The magnitude and phase response of the APF realized from the circuit of Fig. 14.

best knowledge of authors, any first order universal current-mode filter using CFs has not been explicitly reported in the literature earlier. However, no claim is made regarding the DO-CF of Fig. 15 being the best choice to implement the universal filter of Fig. 14; other better DO-CF architectures might be possible to implement the circuits, since the kernel of the configuration of Fig. 14 is independent of the implementation of the DO-CF employed. The circuit of Fig. 14 employs the proposed floating positive VCR of Fig. 1, as the tuning element having equivalent resistance value $R_O = R_{eq}$ as given in Eq. (8). By a straight forward analysis, the various transfer functions of the universal filter of Fig. 14 can be expressed as:

$$\frac{I_{02}^-}{I_{in}} = \frac{-1}{(1 + sR_O C_O)} \quad (15)$$

$$\frac{I_{01}^+}{I_{in}} = \frac{(sR_o C_o)}{(1 + sR_o C_o)} \quad (16)$$

$$\frac{I_{03}}{I_{in}} = \frac{(1 - sR_oC_o)}{(1 + sR_oC_o)} \quad (17)$$

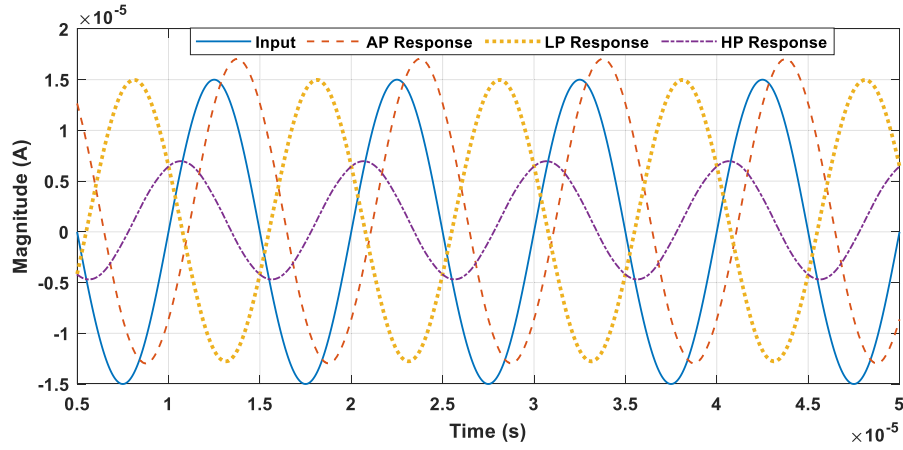


Fig. 18. The transient responses of the first-order voltage-controllable current-mode universal filter.

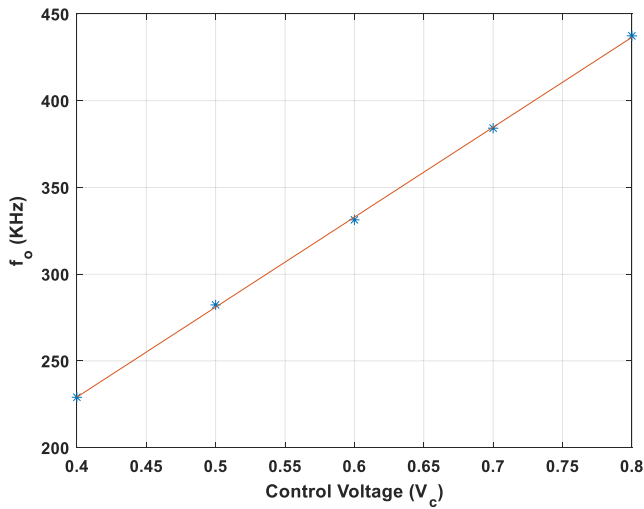


Fig. 19. The variation in f_o with respect to the control voltage V_c .

The circuit, thus, realises an inverting low pass filter at I_{02} , a non-inverting high pass filter at I_{01}^+ and an all-pass filter at I_{03} .

Eq. (17) represents a first order all-pass filter having magnitude as unity whereas the phase can be expressed as:

$$\Phi(\omega) = -2\tan^{-1}(\omega R_O C_O) \quad (18)$$

The cutoff frequency of the low pass and high pass filters, as well as the frequency at which, the phase angle of all-pass filter becomes -90° , is given as

$$f_o = \frac{1}{2\pi R_O C_O} \quad (19)$$

Using Eq. (8) and taking $R_O = R_{eq}$, Eq. (19) can be modified as:

$$f_o = \frac{KV_C}{\pi C_O} \quad (20)$$

From Eq. (20), it is observed that f_o of the three filters is linearly fine-tunable with the control voltage V_c .

To verify the workability of the proposed floating positive VCR of Fig. 1, the circuit was simulated by taking $|V_{DD}| = |V_{SS}| = 0.75$ V and $I_{B1} = 100$ μ A. The W/L of the various MOSFETs were taken as: $(W/L)_1 = (W/L)_2 = (30/0.18)\mu\text{m}$, $(W/L)_3 = (W/L)_4 = (90/0.18)\mu\text{m}$, $(W/L)_5 = (W/L)_6 = (W/L)_{11} = (W/L)_{12} = (W/L)_{13} = (W/L)_{14} = (W/L)_{15} =$

$(12.5/0.18)\mu\text{m}$ and $(W/L)_7 = (W/L)_8 = (W/L)_9 = (W/L)_{10} = (W/L)_{16} = (W/L)_{17} = (W/L)_{18} = (W/L)_{19} = (5/0.18)\mu\text{m}$. The frequency response of low pass filter and high filter for $C_O = 10$ pF and the control voltage taken as 0.6 V is shown in Fig. 16, both of which exhibits a cut off frequency of 331.325 kHz. Moreover, the frequency response (magnitude as well phase) of the first order all-pass filter is found to be as given in Fig. 17, where the magnitude of the all-pass filter is seen to be almost the same when the control voltage is varying from 0.4 V to 0.8 V in steps of 0.1 V whereas, the frequency at which phase angle of the all-pass filter becomes -90° , varies from 229.066 kHz to 437.366 kHz, for V_c varied from 0.4 to 0.8 V as shown in Fig. 17. However, this is further clarified in Fig. 19 which shows variation f_o with respect to change in V_c .

To test the fine tunability aspect of the proposed positive floating VCR of Fig. 1 as well as that of the first order universal current-mode filter of Fig. 18, the filter was designed for a nominal value of $f_o = 250$ kHz, by choosing the external capacitor as $C_O = 13$ pF. The designed value of R_{eq} is found to be 45.49 k Ω , which would be realised at the value of the control voltage around $V_c = 0.6$ V. However, the f_o obtained from the simulation has been found to be 234.96 kHz. The fine tunability aspect was tested by varying V_c from 0.4 V to 0.8 V through which it was found that at the value of $V_c = 0.634$ V, the value of f_o becomes exactly 250 kHz.

To demonstrate that the first order universal filter of Fig. 14 operates well, we show the transient waveforms of the input current I_{in} and three output currents I_{01}^+ , I_{02}^- and I_{03} in Fig. 18. It has been verified that the circuit can handle input currents up to ± 20 μ A, without causing saturation of any of the three output currents thereby confirming that the linearity of operation is maintained till ± 20 μ A.

The results given in Figs. 16-19, thus, validate the proper functioning of the proposed first order universal current-mode filter of Fig. 14 as well as the fine-tunability of f_o (from 229.066 kHz to 437.366 kHz) (as in Fig. 19).

It may be mentioned that while all the three filter functions are available from high output impedance nodes, however, if ideally zero (or very low) input impedance is also desired, one more single output CF can be added at the input.

6. Concluding remarks

In this paper, a CMOS configuration employing eighteen transistors, all operated in saturation, has been presented which can realise a positive or negative electronically fine-tunable linear floating VCR from the same configuration by changing two interconnections therein. The realised resistance value is V_{TH} -independent in both the cases. The operation of proposed circuit has been verified by Cadence SPICE tools

using 0.18 μm CMOS technology parameters. The positive floating VCR has been found to vary from 71.26 k Ω to 35.73k Ω whereas the negative floating VCR varies from -71.06 k Ω to -35.53 k Ω , when the control voltage V_C is changed from 0.4 V to 0.8 V in both the cases. The circuit has been found to exhibit good linearity over the input voltage range of ± 0.8 V. The proposed configuration has low power consumption of 0.30 mW for both positive and negative floating VCRs with DC bias supply of ± 1.25 V. To substantiate the use of the proposed floating VCR, a new first order universal current-mode filter realised with DO-CF has been proposed, which employed the CMOS floating positive VCR, along with only one capacitor as the external passive element. The workability of the first order current-mode universal filter and its tunability have also been verified using Cadence SPICE simulations, which was demonstrated by the presented results.

CRedit authorship contribution statement

Harshmani Yadav: Writing – original draft, Validation, Software, Resources, Investigation, Formal analysis, Data curation. **Urvashi Bansal:** Writing – review & editing, Supervision, Project administration, Formal analysis. **Raj Senani:** Writing – review & editing, Supervision, Project administration, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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